

RAJEEV SHARMA

SOFTWARE ENGINEER | PYTHON • JAVA • DATA & AI

Berlin, Germany | [+49 176 6120 4130](tel:+4917661204130) | rjsharma.ing@gmail.com | rajeev-dev.vercel.app | linkedin.com/in/rjsharma | github.com/Rjsharma95

PROFILE

Computational Engineering graduate with nearly two years of hands-on R&D experience in Python, Java, Linux-based systems, data processing, and machine learning. Built software for technical interfaces, reproducible sensor-data pipelines, and an end-to-end CNN workflow, and independently delivered a responsive React portfolio. Comfortable working across software, data, and embedded environments, with a practical focus on debugging, maintainability, and learning new technologies quickly.

PROFESSIONAL EXPERIENCE

Working Student - Research & Development

11/2024 - 09/2026

SectorCon Ingenieurgesellschaft mbH | Berlin, Germany

- Developed and maintained Python processing pipelines for real medical sensor data, covering preprocessing, segmentation, feature extraction, analysis, and validation.
- Developed Java applications to configure and control SPI interfaces and improved the associated data-conversion workflow for reliable transfer.
- Set up and managed Raspberry Pi systems under Linux; performed remote execution, debugging, troubleshooting, and systematic error analysis across software, sensors, and embedded hardware.
- Built, trained, and evaluated a TensorFlow/Keras CNN for biomedical signal-quality classification using scientific Python libraries.
- Adapted and quantized a trained model for the BrainChip Akida development environment and assessed architectural compatibility and effects on model quality.
- Used Git/GitLab, completed technical work packages independently, and communicated results through documentation, diagrams, reports, and presentations.

BACHELOR THESIS - GRADE 1.3

Signal Quality Analysis of Medical Sensor Signals Using Neural Networks and Implementation on a Neuromorphic Chip

03/2025 - 07/2025

SectorCon Ingenieurgesellschaft mbH / Technische Universität Berlin

- Independently planned and implemented an end-to-end Python workflow from acquisition of real pulse-wave measurements through preprocessing, feature engineering, model training, evaluation, and Edge AI adaptation.
- Converted noisy signals into consistent datasets using FIR filtering, segmentation, Z-score normalization, statistical features, and wavelet-based analysis.
- Designed and iteratively improved a TensorFlow/Keras CNN, evaluated it with multiple classification metrics including ROC/AUC, and compared handcrafted features with deep-learning approaches.
- Modified the model and processing workflow when initial approaches were incompatible with the resource-constrained neuromorphic target environment.

TECHNICAL SKILLS

Programming	Python, Java, SQL, JavaScript, C, Bash, MATLAB, R; object-oriented programming
Backend & Web	Flask, REST API concepts, React, HTML/CSS, Tailwind CSS, Vite; Node.js and Django basics
Data & AI	NumPy, pandas, SciPy, PyWavelets, TensorFlow/Keras, CNNs, preprocessing, feature engineering, signal processing, model evaluation
Systems & Tools	Linux/Unix, Docker, Raspberry Pi, Git, GitHub, GitLab, remote execution and debugging
Databases & Cloud	SQL, Oracle basics, AWS EC2 and Microsoft Azure basics

SELECTED PROJECT

Personal Portfolio Website	2026
React, JavaScript, Tailwind CSS, Vite, Node.js tooling, GitHub	
- Designed and built a responsive portfolio with reusable React components, custom styling, section-based navigation, and mobile layouts. - Implemented interactive UI elements and a working contact flow; used Git/GitHub throughout development and deployed the finished site on Vercel. - Live: rajeev-dev.vercel.app Source: github.com/Rjsharma95	

SELECTED ACADEMIC PROJECTS

Concept Design of an Autonomous Mobile Cleaning Robot	TU Berlin
Interdisciplinary university project	
- Planned a technical concept covering route planning, LiDAR sensing, Raspberry Pi control, drive selection, interfaces, safety requirements, and engineering calculations. - Evaluated system components and documented technical decisions and architecture choices.	
Data Analysis and Anomaly Labelling	TU Berlin
University project with manufacturing data	
Analysed and labelled anomalies in technical manufacturing data and prepared the results in a structured, traceable form.	

EDUCATION

Bachelor of Science in Computational Engineering	2017 - 2025
Technische Universität Berlin Final grade: 2.8	
Focus: programming, computer engineering, artificial intelligence and machine learning, data analysis, and automation engineering.	
Studienkolleg - T-Kurs	2015 - 2016
Technische Universität Berlin	

LANGUAGES

German - Fluent (C1) English - Fluent (C1) Nepali - Native Hindi - Fluent French - Beginner (A1)
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